Oral-Exam 42112, 2025, preparation assignment problem

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May 22, 2026

This is the pre-oral exam assignment for the re-exam of the "Mathematical Programming Modelling" (42112) course. The oral exam takes place for each student either 2/6 2026 or 4/6 2026. This assignment is emailed to all students who have signed up for the re-exam, Friday 22/5. To take the oral exam, the student **has** to hand in two julia programs, with model answers to the two questions, in an email to Thomas Stidsen: thst@dtu.dk . The deadline for hand in is Friday 29/5, kl. 23.59. Half of the oral exam will be questions to these solutions.

Production planning

The company Single Item Production (SIP) contacts you and ask for help with their production planning problem. SIP has specialized in single item production using 3D printing, followed by polish and painting. Customers order on-line and SIP simply needs to come up with a production plan for the three proceses, which are all automated. SIP has just one worker, the owner, who moves the items between the machines. The customers, bid for the production, by uploading the 3D printing plans, polish request and colour request, and what they are prepared to pay. Each day, SIP (the owner) selects the production of the day and produces these items. The customers are **not** guaranteed to get their items produced (they may have offered a too low a pay). There are I item orders from B buyers, some of the buyers want to buy several items. SIP want you to setup a production planning model, where you select the products to produce in a day. The production limits is given by the time required for the item for each machine. The production time available for each mashine is 60 minutes for 8 hours. We ignore the switching time between mashines an the starting times (i.e. assume we can polish from the first minute of the day). For each item we have the following information:

- $Price_i$: The price the buyer is ready to pay for item i, in Euro
- $PrintingTime_i$: The number of minutes item i requires in the 3D printer
- $PolishingTime_i$: The number of minutes item i requires in the polishing mashine
- $PaintTime_i$: The number of minutes item i requires in the painting mashine
- $ByuerId_i$: Which of the B buyers wants to buy this item

The data are in the SIP_data.jl file.

Assignment 1: Make a mathematical model which select the most profitable items for production.

Assignment 2: Transport of the items costs a certain amount, and SIP covers the cost. There is a unit cost of 50 DKR for transport to each buyer. Maximize the total profit, taking into account the transport costs.